

Racial Segregation and Black-White Longevity Disparities

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1 Abstract

Residential, racial segregation defines the life chances of Black Americans and White Americans, but in potentially different ways. In contemporary and historical perspective, Whites have resisted residential integration, despite the fact that desegregation in educational and institutional settings had virtually no impact on Whites' outcomes. A central pillar of this resistance is the perception that Black-White race relations are a zero sum. Despite implicit assumptions that segregation advantages White Americans, this assumption bears empirical scrutiny. To provide formally test whether and how much segregation advantages Whites, I study the effects of segregation on the longevity of Black and White Americans using data from the 1940 Full Count Census linked to Numident death records and plausibly exogenous variation in the density and Federal revenue share of local governments. I show that racial segregation reduces the longevity of Black and White Americans, albeit at different magnitudes. White aversion to integration — rather than advantaging Whites — actually harms their health. Rather than resulting in a zero-sum relationship, policies promoting residential integration would substantially improve Black and White health.

2 Introduction

Despite political and social tendencies toward multiculturalism in the United States, White Americans exhibit pronounced resistance to racial integration in residential, social, and institutional spaces (Frey and Farley 1996; Johnson 2019; Rich n.d.; ?; ?). The persistence of Black-White residential, racial segregation (Elbers 2021) represents one of the most visible and salient dimensions of Black-White inequality (?) and is thought to shape inequalities in economic and physical wellbeing (Massey and Denton 2003). A central pillar of White resistance to racial integration, is the perception that Whites will lose advantages afforded to them by existing racial structures. Extensive research on Whites' attitudes toward race relations more generally (Gilens 1996; Quillian 1995, 1996; Warren and Valentino 2025; ?) and residential integration with Black Americans (Bobo and Zubrinsky 1996) shows that they view progress toward racial equality as a zero-sum relationship, wherein reducing Black disadvantage comes at a cost to Whites. While research on the ways that racism affects White Americans is not new (?) the majority of this research has focused on Whites' beliefs about racial integration (Bobo and Zubrinsky 1996) and racialized policies (Gilens 1996, 2000; Quillian 1995), rather than whether and how much segregation advantages Whites.

The overwhelming inertia of this literature is expect that the effects of racial segregation are advantageous for White Americans, however little empirical evidence has addressed this question (Light and Thomas 2019). Despite this, zero sum framework is a pervasive and has historically undermined political efforts toward broad based investment in the social safety net and public services that would benefit all Americans (Blumer 1958; ?). However, research casts doubt on the veracity of the zero sum perspective. In some cases, structural racial inequality has been shown to have negative consequences for Whites' health (?; ?) and the

intergenerational mobility of White children (?). In the context of school desegregation, research generally shows that desegregation had no effect on White children's outcomes despite transformative change on the academic achievement of Black children ([Anstreicher, Fletcher, and Thompson 2022](#); [Johnson 2019](#)). At the macro level, residential segregation undermines the political efficacy of politicians and policies advocating for redistributive policies (?) and is associated with reductions in spending on public services and infrastructure (?) that affect all residents. Despite the fact that the zero sum framework and the assumption that Whites' benefit from segregation are deployed frequently in research and popular narratives, this assumption bears empirical scrutiny. On one hand there are credible theoretical and empirical arguments that segregation shields Whites from disadvantage ([Massey and Denton 2003](#)), however there are also important system-wide consequences related to institutional and political environments that may disadvantage Whites in segregated spaces.

This goal of this paper is to evaluate the empirical basis for zero sum framework by evaluating the assumption that structural racial inequality advantages White Americans, using the case of residential segregation and Black-White longevity disparities. Population health disparities between Black and White Americans are stark, in both contemporary and historical settings, which serves as an ideal measure life chances. I employ longevity as a measure because it is both an important dimension of population health and an axis of social inequality between Black and White Americans ([Maralani and Portier 2021](#); [Wrigley-Field 2024](#)). Beyond health status, longevity disparities have far-reaching implications for social networks, collective memory, culture, political participation, and social movements ([Wrigley-Field 2025](#)) Longevity thus is an omnibus measure of non-economic wellbeing and an important descriptor of the summation of life chances ([Montez et al. 2020](#); ?). Analytically, I study the effects of segregation on the longevity of Black and White Americans born in the early 20th Century (1905-1920), using data from the 1940 Census linked to CenSoc Numident

death records ($N = 2,157,524$) and an instrumental variable strategy leveraging variation in the density of governments and their revenue share in U.S. counties to instrument for racial segregation ([Cutler and Glaeser 1997](#)).

Results indicate that segregation decreases Black longevity by **2.8 years** and decreases White longevity by **.8 years years overall**. The negative effects of segregation for Black Americans are especially pronounced in the South, whereas effects are negative but insignificant outside of the South. These results for Black Americans are consistent with literature on the effects of the Great Migration suggesting unclear or even negative effects of leaving the South ([Black et al. 2015](#)). For White Americans, the effects of segregation within and outside the south are similar. These results emphasize the importance of the effects of structural racism on the health of both Black and White Americans, that while segregation harms the longevity of Black Americans, it also harms the health of White Americans. Consistent with arguments counter to a zero sum framing of Black-White inequality, racial segregation — rather than advantaging Whites — reduces their lifespans. Policies promoting integration would substantially improve Black health outcomes while also benefiting White Americans. In the context of a Black-White life expectancy gap of 6.3 years in 2000 ([Arias n.d.](#)), these results imply that moving from a perfectly segregated to perfectly integrated county would narrow the gap to 3.7 years $((77.4 + .8) - (71.7 + 2.8)) = 3.7$ or about half of the observed gap.

3 Background

3.1 The Zero Sum Framework and the Consequences of Segregation for White Americans

Historically, residential segregation has emerged as the “‘structural linchpin’ of race relations in America,” ([Bobo and Zubrinsky 1996](#) : 883) and represents one of the most consequential and visible dimensions

of Black-White inequality in the U.S. Sociological research has long documented the significance of the differential residential arrangements of Black and White Americans (?). The residential separation between Black and White Americans is thought to be a driving force behind persistent Black-White inequalities both within and across generations (Sharkey 2013; ?). One barrier to declines in segregation is White resistance to residential integration through white flight (Crowder and South 2008; Frey 1979; Logan, Zhang, and Oakley 2017), avoidance from cities (Rich n.d.), suburbanization (Owens and Rich 2023), and resistance to school desegregation (Anstreicher, Fletcher, and Thompson n.d.; Johnson 2019).

Established sociological explanations for the what drives Whites to resist progress toward racial equality argue that these beliefs stem from perceptions of out group threats to their advantaged position (Blumer 1958; Wetts and Willer 2018). Public opinion research shows that White Americans view progress toward racial equality with Black Americans as a zero sum relationship, where the advancement of Black Americans necessarily comes at a cost to Whites (Bobo and Zubrinsky 1996; Quillian 1995, 1996). The racial politics of Black and White Americans have important political consequences, particularly in the context of redistributive policy. For instance, Whites' views about welfare policy are highly racialized (Gilens 1996) and leading Whites to disapprove of these policies (Gilens 2000). At a macro level, McGhee (2021) argues that political efforts toward working class policies more generally are been undermined by racialized messaging to Whites that plays into zero sum thinking (Wetts and Willer 2018). Evidence also indicates that policies that solidarity-building policies, such as unions, lead to declines in racial resentment (Frymer and Grumbach 2021).

While past work directs out understanding of Whites' beliefs about threats to their status in the racial hierarchy, research has neglected to empirically examine the empirical basis for the zero sum framework. Despite the fact that responses to integration suggest that Whites reap significant material advantages to segregation, the evidentiary basis for this is mixed. In fact, in past studies there is scope for segregation to

have either muted or deleterious effects on White outcomes. In *Black Reconstruction in America*, DuBois (1935) argues that Whites are paid psychological wage paid to offset the poor material circumstances faced of Whites in similar class position to Black Americans. This perspective proposes that Whites are actually disadvantaged by segregation in ways that stem from its structural nature and system-wide implications of living in a segregated place.

In research on the effects of school desegregation orders on children's outcomes, empirical research demonstrates major gains for Black children in the form of human capital, test scores, and labor market outcomes but no effects on White children's achievement (Anstreicher et al. 2022; Johnson 2019). In other words, school segregation is very bad for Black children's outcomes but indifferent to Whites. Additionally, popular narratives attributing school desegregation to White population loss have been shown to be muted (Rich n.d.)

In the empirical literature, segregation may affect Whites indirectly, by increasing their housing financial burden because of the cost-premium in majority white neighborhoods (Daniels 1975). Segregation has also been shown to reduce the inter generational mobility of White children (Chyn, Haggag, and Stuart 2023) and undermine the political efficacy of liberal political candidates advocating for policies targeted and working class issues that would benefit individuals regardless of race (Ananat and Washington 2009). Segregation also has far reaching consequences for the quality and funding of public goods and services accessed by all. Chyn et al. (2023) demonstrates that segregation decreases spending on public safety, welfare, infrastructure, and health services that are critical for individual and population health and economic wellbeing,

Finally, segregation severely limits the potential for interracial friendships (Mouw and Entwisle 2006), marriages (Goldman, Gracie, and Porter n.d.), and diverse social networks to be formed, which have implications for social capital. Despite compelling and varied pathways through which segregation could

affect White Americans, there is still significant scope for ambiguity.

3.2 Prior Work on Racial Segregation and Health

Past work has consistently established that segregation is associated with negative health outcomes for Black Americans. Using an instrumental variable strategy, Austin, Harper, and Strumpf (2016) found that segregation decreased the birthweight of Black children by 2.8 grams and decreased the birthweight of White children by .63 grams. Similarly, Vu, Green, and Swan (2024) also study effects on birthweight and finds negative impacts of segregation on the birthweight of Black children. Their results indicate that the primary mechanisms that drive these outcomes arise from disparities in access to prenatal care for Black women, structural racism, and food insecurity. In contrast, Smith (2024) found that in some cases, segregation may be beneficial for Black mental health, though these processes may be distinct from those affecting physical health and longevity. Finally, (?) that shows that segregation is responsible for a 16 year increase in the Black-White life expectancy gap for men and a 5 year increase for women, based on period life expectancy estimates between 1990-2019.

While the bulk of studies rightly focus on health outcomes for Black Americans, results for White Americans are somewhat more ambiguous. Vu et al. (2024) for instance found that there was no effect of segregation on the birthweight of non-Hispanic White children but Austin et al. (2016) found that segregation decreased White childrens' birthweight significantly. Logan and Parman (2018) found that in certain contexts, segregation was associated with greater mortality for White Americans that is corroborated in a similar study by Karbeah and Hacker (2023) who found that segregation was associated greater mortality among Whites. One important note about Logan and Parman (2018) and Karbeah and Hacker (2023) is that they use a micro-level measure of segregation based on the racial composition of one's neighbors, which is not directly comparable to macro-level measures.

While studies have documenting the effects of segregation on on health outcomes has been extensively documented, results across studies are far from conclusive and important methodological questions remain about whether these associations are causal. For example it could be that patterns of residential segregation and individual health and longevity are driven due to unobserved heterogeneity, preferences, or unobserved neighborhood characteristics. Moreover, work by Montez et al. (2020) demonstrates that there are many aspects of places, such as the policy context, that are bad for population health that may be associated with levels of segregation. Thus, correlational analyses cannot distinguish whether health outcomes are driven by segregation or other factors. This study makes important contributions to the empirical literature by studying the causal effect of segregation on individual-level longevity of Black and White Americans, using rich, nationally representative full count Census data.

3.3 Theory Linking Racial Segregation to Health

County-Level Mechanisms

To date, the majority of studies linking segregation to health outcomes have focused on between neighborhood mechanisms, but focusing on neighborhoods neglects important interconnected social, legal, political, and educational factors at the macro-level that affect longevity (Brown and Homan 2024:2). (Brown and Homan 2024; Montez et al. 2020). The role of administrative units, such as the county or state, are important from a public policy perspective, because they are intrinsically linked to important political, legal, and institutional processes that affect individuals living within these boundaries. Conceptually, focusing on the county-level as the unit of analysis is congruent with segregation theory and measurement, which refer to the distributions of populations at smaller geographies within a larger geography (Duncan and Duncan 1955). Studying segregation at smaller geographies may be sensitive to neighborhood change over time as census tracts change over time and their populations shift dynamically. Despite the fact that individuals may move

between neighborhoods, the experience and effects of segregation are shaped by the different populations living in inter-dependent neighborhoods. While micro-level analyses capture how proximate neighborhood effects shaped by segregation affect individual outcomes, county-level analyses captures the average effect of segregation at a structural level irrespective of neighborhood differences.

At the county level, there are a variety of theoretically relevant and compelling pathways that may shape individual health outcomes. Figure 1, describes a stylized empirical diagram relating segregation to longevity, county-level and neighborhood pathways, and relevant confounding processes that are described in this section. Counties that are more segregated have are more likely to have decreased funding for public goods and public opinion in these places indicates decreased support for redistributive policies that could boost the economic conditions of working class populations ([Chyn et al. 2023](#); [McGhee 2021](#)). More generally ([?](#)), [Gilens \(2000\)](#), and [McGhee \(2021\)](#) hypothesize the common factor of these processes is greater racial resentment in more segregated counties. Through this mechanism, greater aversion to public policy that would likely improve the economic conditions of low-income Black Americans and decreased government support would likely have differential impacts on economic and material resources that have the potential to affect low-income populations more broadly, regardless of race.

[Figure 1 Here]

County-level mechanisms also work through the criminal legal system to affect individual outcomes through institutional and governmental mechanisms that are shaped by historical racial inequality ([Brown and Homan 2024](#); [Ray 2023](#); [Williams and Collins 2001](#)). Compared to any other sex-age-race group, Black men and women are at the highest risk of experiencing police violence ([Edwards, Lee, and Esposito 2019](#)). [Bell \(2020\)](#) demonstrates that patterns of aggressive policing and punitive criminal justice policy contexts stem from high levels of segregation. In this way policing is tied to racial composition and the differing residential

arrangements between White and Black Americans. Exposure to aggressive policing in segregated areas may plausibly affect both Black and White mortality through processes of arrest and incarceration. Additionally, Light and Thomas (2019) show that the segregation also creates racialized inequalities in homicide risk both through increasing Black risk, but also mechanically decreasing White Americans' risk of victimization because of residential separation. Beyond extreme outcomes like police use of force, exposure to aggressive policing leads to a higher likelihood of criminal justice contact, which has been linked to negative population health outcomes for Black Americans (Cao, Esposito, and Lee 2025; Lee 2024).

County-level effects of segregation are also shaped by historical processes and long-run patterns of racism and stratification (Graetz and Esposito 2023; Rothstein 2018). Ananat (2011) demonstrates that, in Northern metropolitan areas, segregation led to an increase in poverty for Black Americans and decreased poverty for White Americans. Additionally, segregation increased Black-White income gaps. In contemporary contexts, Graetz and Esposito (2023) show that HOLC redlining maps, which were developed for the sole purpose of maintaining segregation, created greater economic isolation and devaluation of property values in mapped areas that have proximate implications for segregation and health outcomes through homeownership (Breen 2024). At a broad level, research on the historical legacy of segregation suggests that even in an era of decline (Cutler, Glaeser, and Vigdor 1999), patterns of segregation and racial inequality have reverberating effects over time that are important for present day outcomes.

Neighborhood-Level Mechanisms

County-level processes also have downstream effects on within-county neighborhood inequalities that shape health. In places with high levels of segregation, structural forces and individual residential mobility patterns shape residential decisions that are associated with social and environmental disadvantages and unequal opportunities across neighborhoods that stratify the neighborhood conditions of Black and White individuals

([Wilson 1987](#)).

Massey and Denton ([2003](#)) argue that segregation ultimately leads to greater concentrations of poverty, which is consequential for both economic mobility and health ([Aber et al. 1997](#)). Poverty affects health through mortality directly, but also through indirect pathways that influence how health status changes over the life course through exposure to poor residential conditions, resource deprivation, and increased exposure to illness. Moreover, the levels, timing, and duration of poverty are also consequential, particularly for health in adolescence ([Brooks-Gunn and Duncan 1997](#); [Chetty, Hendren, and Katz 2016](#); [Sharkey 2013](#); [Sharkey and Elwert 2011](#)). Segregation leads to higher spatial concentrations of poverty and income segregation within racial groups ([Quillian 2012](#)), that overlap with Black and White residential patterns. Racially stratified exposure to poverty has both direct effects on Black American’s health outcomes by reducing educational and economic opportunities and also indirectly benefits White Americans’ health by reducing their exposure ([Ananat 2011](#); [Williams, Lawrence, and Davis 2019](#)).

Environmental characteristics of neighborhoods are also an important driver of health ([Williams and Collins 2001](#)) shaped by both exposure as well as ultimate political processes and public policies that allowed pollution in poor and Black areas ([Elliott and Frickel 2013](#) : 20). Black neighborhoods are frequently exposed to hazardous environmental conditions and among the highest risk of being exposed to air pollution ([Lopez 2002](#)). Black and low-income populations are also at higher risk of exposure to toxic chemicals and metals such as lead ([Wodtke, Ramaj, and Schachner 2022](#)) which have direct and indirect effects on health and cognitive ability. More generally, these exposures reflect a geography of hazardous environmental exposures is racially and economically stratified, with polluting industries affecting both current sites of production as well as leaving behind remnants of emissions and toxins in disproportionately minority neighborhoods ([Elliott and Frickel 2013](#): 6).

The neighborhood context also significantly shapes the educational opportunities and school quality of Black and White children (Wodtke et al. 2023) both through racially stratified education opportunities and school funding is tied to homeownership and property taxes. Black and White students also attend segregated schools that are shaped by the demographics of their neighborhoods and correlated with educational achievement over the life course (Goldsmith 2009). A prior study by Card and Rothstein (2007) found that all else equal, segregation was responsible for a quarter of the Black-White gap in Scholastic scores. Those who attend schools with greater resources and funding frequently concentrated within White neighborhoods, benefit from the racially stratified geography of educational opportunity.

3.4 The 1940 Context

There are a few features of the data used in this study as well as the observed period of segregation that are important to contextualize when interpreting estimates in this study. The birth cohorts used in this study from 1905-19xx survived exceptional economic hardship from the Great Depression as well as the 1918 Flu Pandemic and were members of the earliest cohorts of Black migrants from the South to the North. These circumstances reflect important selectivity that must be taken into account when situating these estimates within the literature on segregation and health.

The purpose of using the 1940 Census data for analysis are to identify individuals who were part of the 1904-1918 birth cohorts who likely complete their mortality window within the 1988-2005 time frame that the Social Security Administration public data is available for (Breen 2024; ?). These birth cohorts are also socially and historically significant. Members of these birth cohorts are situated in a historical period marked by intense hardship followed by prosperity. These cohorts survived the first and second World War as well as the 1918 Spanish Flu which also had distinct racial disparities (?). During the 1920s and 1930s, these birth cohorts weathered the great depression, where unemployment reached highs of 24 percent (?). Amid these

major changes to American Life were the prosperity that followed these material hardships, spurred in large part by expanded opportunity resulting from the New Deal introduced by President Franklin D. Roosevelt. At the same time as these changes happened, the Great Migration of Black Americans from the South to the North that occurred between 1910 and 1970 led to a restructuring of the landscape of opportunity for Black Americans in the U.S. as reshaped race relations in complex ways (?).

The time period of life that I observed segregation is also important to consider when interpreting estimates. I examine exposure to segregation that mid-to-late life, obtained and calculated based on the county individuals resided at death. This decision is to maximize geographic coverage within and outside the South to better reflect contemporary distributions of Black and White Americans. Prior to the great migration [XXXX percent of Black Americans lived in the North].

Research on the life-course dynamics of migration propensity (?) suggests that individuals are less likely to migrate after age 40 years old and when they do it is usually short distances (?; ?). Additionally, research on aging in place shows that older adults increasingly reside in the same neighborhoods and residential contexts for 30 or more years preceding death (?). For these reasons, examining exposure to segregation in the county of death represents a wider window of exposure stretching back to mid-life. It is also true that prior to the period of segregation I observe, average segregation levels of segregation in counties in my analytic sample were higher than the observation period. Coupled with this trend, work on neighborhood attainment and residential mobility show that the modal neighborhood transition is to one of similar characteristics (?). Thus, I conclude that estimates of exposure to segregation experienced in mid-to-late life, net of birth place characteristics, proxy for life course segregation exposure that are likely attenuated from the levels experienced earlier in life.

3.5 The Present Study

The goal of this study is estimate the causal effect of segregation on the longevity of Black and White American, with the goal of understanding whether Whites’ benefit from segregation resulting in a zero sum. Or, whether segregation disadvantages both Black and White Americans, implying that declines in segregation would lead to longevity dividends. Longevity, defined as the number of years one lives conditional on surviving to 65 years old, is an important omnibus measure of health and life chances ([Breen 2024](#); [Weber 1922](#)). To address this question, I combine linked data from the 1940 Census with data on municipal and township governments to instrument for residential segregation to identify a causal effect of segregation on longevity.

4 Data

4.1 Data Sources

To answer my research questions, I leverage data on 1905-1920 birth cohorts of persons xx-xx years old from the 1940 Full Count United States Census. I link these data with Numident death records and the Supplementary Geographic File provided by the Berkeley CenSoc project that combines geographically linked death records from the Social Security Administration (SSA) (?).

The geographic data in the Numident Geographic Supplement cover deaths occurring between 1988-2005 and reliably cover deaths over the age of 65 during this period. The geographic specificity and rich covariates, harmonized over time, make these ideal for analyses of longevity. I focus on the 1904-1924 birth cohorts for two reasons.

First, their birth years and the end year of the mortality observation window allow for likely complete observation of mortality that is not biased by missing the oldest old. In other words, those born in 1905 would

be 100 years old in 2005, making reasonable to assume that there is not substantial missing, right-censored data on individuals who are still alive. Second, those aged 22-36 likely have completed the majority of their formal education, which allows me to rule out the possibility of confounding related to individuals who are still in the process of completing education (?).

To measure segregation, I use the dissimilarity index, a measure that describes how evenly Black and White residents are distributed in sub-geographies within a county relative to the county-level racial composition, which has been used widely in empirical research ([Duncan and Duncan 1955](#)). As mentioned previously, I calculate the dissimilarity index for the county an individual resided in at the time of their death. The dissimilarity index is interpreted as the share of Black and White individuals that would have to move for the racial composition of neighborhoods to be the same as the racial composition of the county. The dissimilarity index is calculated formally as:

$$D = \frac{1}{2} \sum_i^N \left| \frac{b_i}{B} - \frac{w_i}{W} \right| \times 100$$

Where N is the number of counties, b_i and w_i are the proportion of Black and White residents in the i th census tract and B and W are the proportion of Black and White residents in the county. The 1940 Census also includes data on county of residence at birth which I compare to county at death to determine whether individuals were movers or stayers. The dissimilarities are calculated from decennial Censuses 1980, 1990 and 2000, which correspond to the year that they passed.

My final analytic sample consists of ($N = 2,157,524$) individuals residing in 1,799 counties at their time of death over three decades from 1980-2000. These counties are then linked to 1967 Census of Governments data on the number of municipal and township governments and their Federal Government revenue share from the

Government Finance Database (Pierson, Hand, and Thompson 2015) that I use for identification. The focal filtering criteria that I use is coverage of my instrument and complete individual cases, and Table (xxxx) describes the differences between non-sample versus sample counties, that appear to be broadly representative. These data provide geographic coverage in all regions of the U.S. and the average dissimilarity ranges from (0-.94). While the majority of individuals in the sample reside outside the South, counts are highly distinct by race where (62,291) Black individuals reside within the South compared to (46,390) outside. On the other hand, the majority of Whites reside outside the South (1,492,671) compared to (556,172) within the South. On average, sample counties in the South had a population size of 103,553 that was 17.9 percent Black compared to 149,111 outside the South that was 2.7 percent Black.

Covariates and Adjustment for Double Truncation

In addition to the instrumental variable strategy, I control for a rich set of individual and contextual covariates to adjust for any residual confounding. In fully specified models, I control for sex, education, marital status, employment status, whether an individual migrated, whether they reside in the South, urban-rural fixed effects, birth year and state fixed effects, period fixed effects, and occupation fixed effects.

Birth year fixed effects are integral to reducing bias in mortality analysis with truncated data, where individuals of varying birth cohorts may have differing numbers of observations covered by the death window. It is well known that the NUMIDENT data suffer from the problem of double truncation that refers to the left and right censoring of the data (?). (?) show that when not accounted for, doubly truncated observation windows result in biased estimates of longevity disparities toward 0 and mean age at death is higher. This means that, for those born in 1904, we observe them between the ages of 85-100 years old and for those born in 1920 we observe them between the ages of 68-85 years old.

It is also possible that there is important unobserved heterogeneity in early life experiences and residential

environments common to one’s place of birth that could bias estimates of segregation on longevity. To adjust for early life characteristics of individual birth places, I include birth county fixed effects from the geographic supplement in the NUMIDENT to address early life exposures.

Despite the strengths of these data there are important limitations. In terms of record matching algorithms, matches between the 1940 Census and the NUMIDENT may favor more advantaged Black Americans, resulting in estimates of the effect of segregation that are conservative. These analyses are also conditional on having no missing data in the NUMIDENT geographic supplement, which adds an additional layer of selectivity. Substantively, doubly truncated data is also a limitation because, while it is possible to recover estimates of longevity disparities, the narrow time window limits understanding whether longevity outcomes are primarily caused by earlier or later exposures to segregation (?). To address these representation issues, I use post-stratification weights developed by (?) that re-weight CenSoc NUMIDENT counts to year-specific distributions of deaths to adjust for possible selection. Table 7 in the appendix compares the analytic sample used in this paper to the full NUMIDENT population and these populations are, on average, comparable though my analytic sample is positively educationally selected and live in slightly more segregated counties.

5 Methods

5.1 Identification Strategy

Research has established that there are significant barriers to drawing causal inferences from observational data on the relationship between segregation and longevity. In particular, recent work on the place-level correlates of population health disparities show that there are many features of state-level policies contexts, which necessarily affect counties, that adversely shape health (Montez et al. 2020), leading simple associations between county-level segregation and longevity to be biased.

A leading strategy to alleviate concerns of endogeneity and unobserved heterogeneity is to find an instrumental variable that induces exogenous variation in segregation, but does not plausibly affect the outcome of interest, except through segregation. To identify a causal effect of segregation on longevity, I use an instrumental variable strategy first developed by Cutler and Glaeser (1997) and employed subsequently by Light and Thomas (2019) that leverages variation in the number of municipal and township governments and the state-level federal government revenue share these governments receive as instruments.

These instruments are thought to induce segregation through a Tiebout mechanism: “when there are more local governments, tax rates and service provision will vary more within an area, and thus the desire for sorting will increase,” (Cutler and Glaeser 1997:848). This quasi-randomly induced sorting is thought to mechanically lead to more segregation that is not related to empirical outcomes of interests (e.g., income, test scores, crime). In this paper, I extend this strategy to study health and longevity. While it may be the case that the number of governments and quality of services typical to each government may be correlated with health, I rely on the 1967 Census of Governments which is sufficiently temporally prior to the study period of 1980-2000. As noted by Cutler and Glaeser (1997) and Light and Thomas (2019), the correlation in the number of governments over time is very high at over .98 (Cutler and Glaeser 1997:848). Furthermore, these government patterns exhibit a high degree of correlation across an even longer time horizon, stretching back to 1787 in some cases, suggesting that their influence on sorting patterns occurred sufficiently prior to the segregation studied in this paper. For the purposes of this paper, information from the Census of Governments are obtained from the Government Finance Database (cite) and merged to counties to obtain aggregated information. In the regression setting, I follow Cutler and Glaeser (1997) and include the log of the number of governments and the state-level share of intergovernmental revenue to purge local endogenous factors.¹

¹The first stage is estimated as $D = \alpha + \ln(n_governments) + revenue_share + \epsilon$.

When evaluating the plausibility and relevance of this instrument, it is important to do so in the context of IV assumptions. (1) the instrument is exogenous and relevant (this cannot be tested formally), (2) strength of the first stage relationship (the IV must substantially affect the endogenous variable), and (3) monotonicity (the IV only influences the level of segregation in a county in one direction).

First, the assumption that (1) these instruments impact longevity only through segregation. It is possible that the number of governments could affect other population outcomes (e.g., racial composition or population size) that may affect longevity and segregation through unobserved pathways. (2) In Figure X and table X, I demonstrate that association between the instruments and segregation is highly predictive. The first-stage F-statistic tests that test the strength of the instrument and segregation and indicate a strong and significant F-test of **XXXX** (Table 2 in Appendix) (?). and (3) it is unlikely that these instruments result in sorting that reduced the level of segregation in a city though it is important to note that the interpretation of monotonicity in a setting where the instrument does not take the form of an offer that an individual can respond to is not straightforward ([Morgan and Winship 2014](#)).

Because there are more instruments than endogenous regressors, I conduct a Sargan overidentification test that checks whether the instruments are appropriate and identify the same parameters [**Get Cite**], and this test cannot reject the hypothesis that that the instruments should be included ($p = .xx$).

Provided that these assumptions are satisfied, IV estimates of a regression of longevity on segregation are interpreted as the local average treatment effect (LATE) (?), or the treatment effect amongst those living in counties where segregation increased as a result of greater sorting from increased government density Despite limitations, to date, these instruments has been used effectively in prior studies to examine how segregation effects multiple outcomes ([Cutler and Glaeser 1997](#); [Light and Thomas 2019](#)) In the context of trying to recover estimates as close to causal as possible, randomized experiments manipulating individual residential

contexts are not feasible nor ethical.

5.2 Estimation

Empirically, I estimate the relationship between segregation and age at death, conditional on living to 65, in the county of residence at death. I estimate the following regression via two-stage least squares estimation, stratified by racial group. In all models, I include robust standard errors to account for heteroskedasticity. Standard errors clustered at the county-decade level to account for spatial and temporal variation yield similar results.

$$Y_i = \beta_0 + \beta_{Segregation} + \gamma_{birth-year} + \lambda_{birth-county} + \beta X_{controls} + \epsilon_i$$

In summary, this empirical strategy addresses critical endogeneity and selection issues in estimating the relationship between segregation and longevity, stemming from individual preferences and structural constraints and confounding the relationship. Based on the assumption that the instrument only influences longevity through segregation, estimates can be interpreted as causal. Moreover, interpreting these results in context, the nature of the selectivity of the 1905-1920 birth cohorts, the selectivity of examining life expectancy over 65, and the selectivity in how individuals are matched from the 1940 Census to NUMIDENT point toward these estimates being conservative and attenuated. For Black Americans, in particular, descriptive statistics indicate that this sample is advantaged along death age and education, suggesting that effects of segregation are also likely attenuated from what the total population effects would be. In an ideal world with complete data coverage, estimates of the effect of segregation on longevity would likely be higher than the results presented in this paper.

6 Results

7 Discussion

Answering this question thus helps clarify the roots of White aversion to integration but also enriches scholarly understanding of the distributional consequences of racial inequality.

Understanding of the material consequences of segregation for Whites is important because it otherwise limits our ability to understand the distributional consequences of racial inequality and how they shape actors' decisions in support or opposition.

More generally these results limit our scholarly understanding on the distributional consequences of segregation. While subtle, establishing this distinction is particularly important for making policy to redress profound racial inequalities in health and longevity, because effective policies require broad appeal including a significant share of the voting bloc, which is White.

which has implications for policy aimed at redressing racial inequalities in health.

a zero-sum hypothesis argues that segregation materially advantages Whites, a Du Boisian hypothesis argues that segregation affords psychological advantages despite neutral or even negative effects of racial inequality for Whites.

8 Tables and Figures

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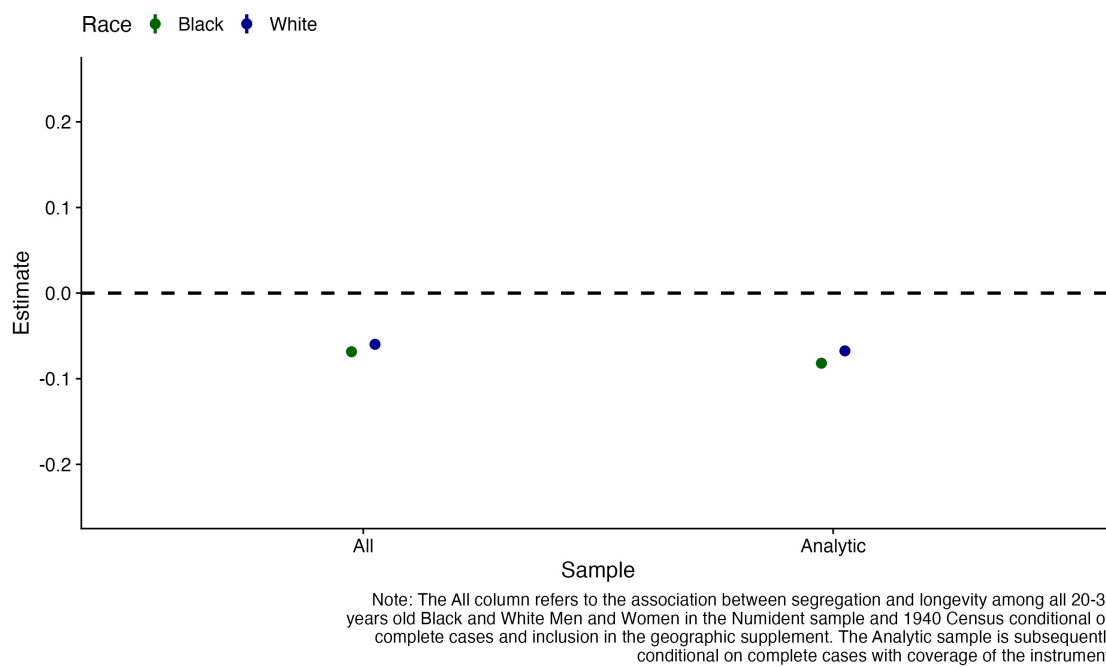


Figure 1: Association Between Seg and Longevity by Sample

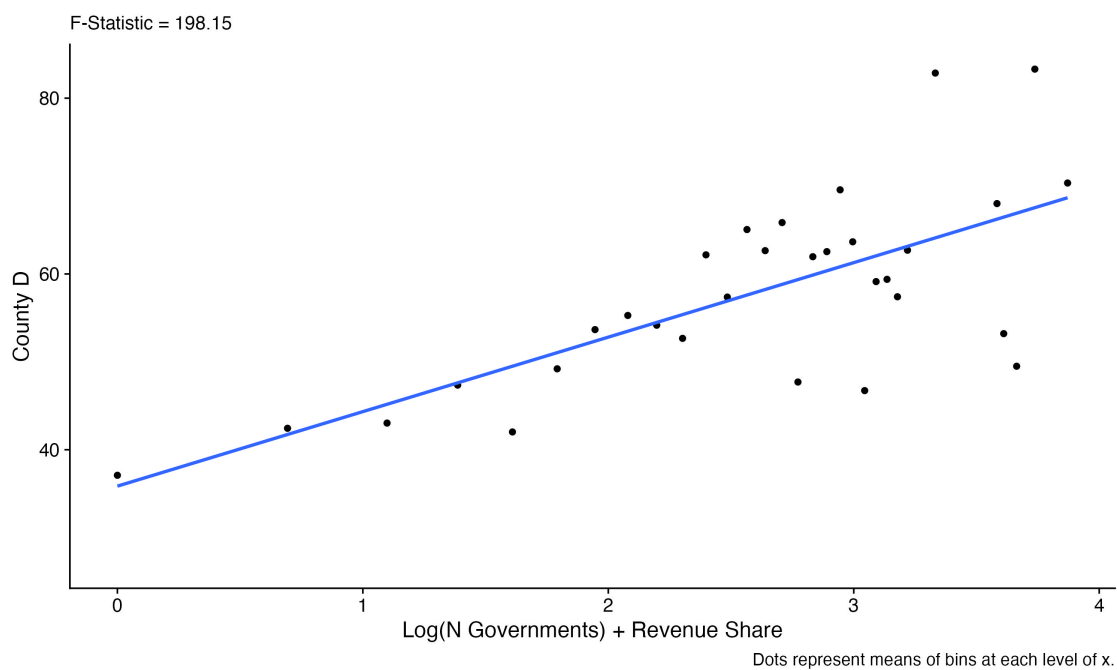


Figure 2: First Stage

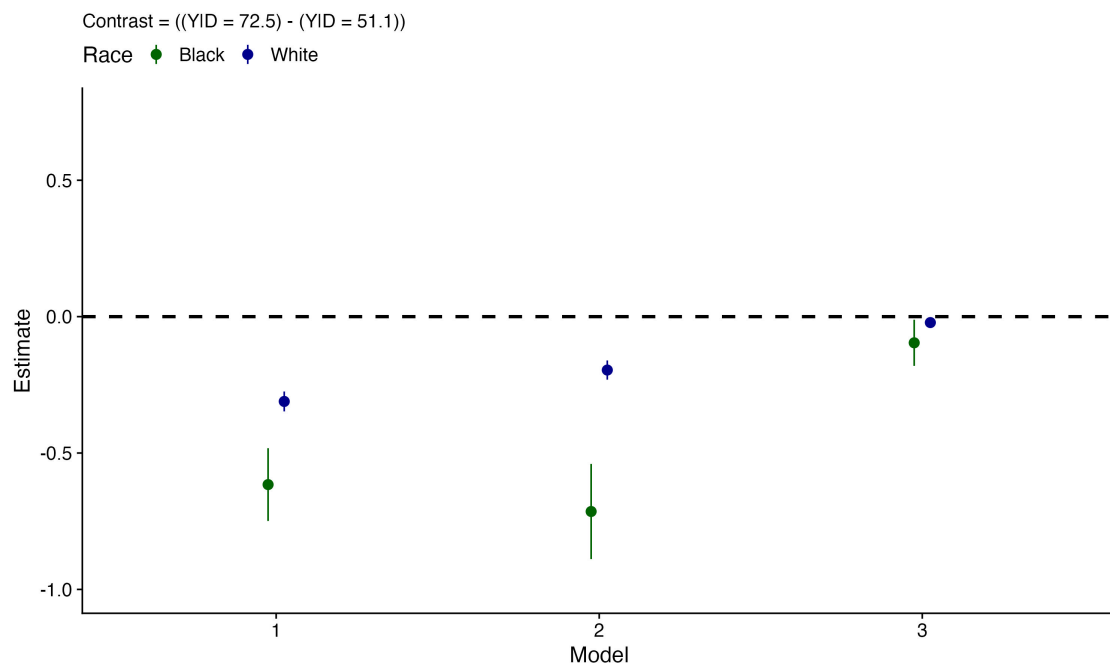


Figure 3:

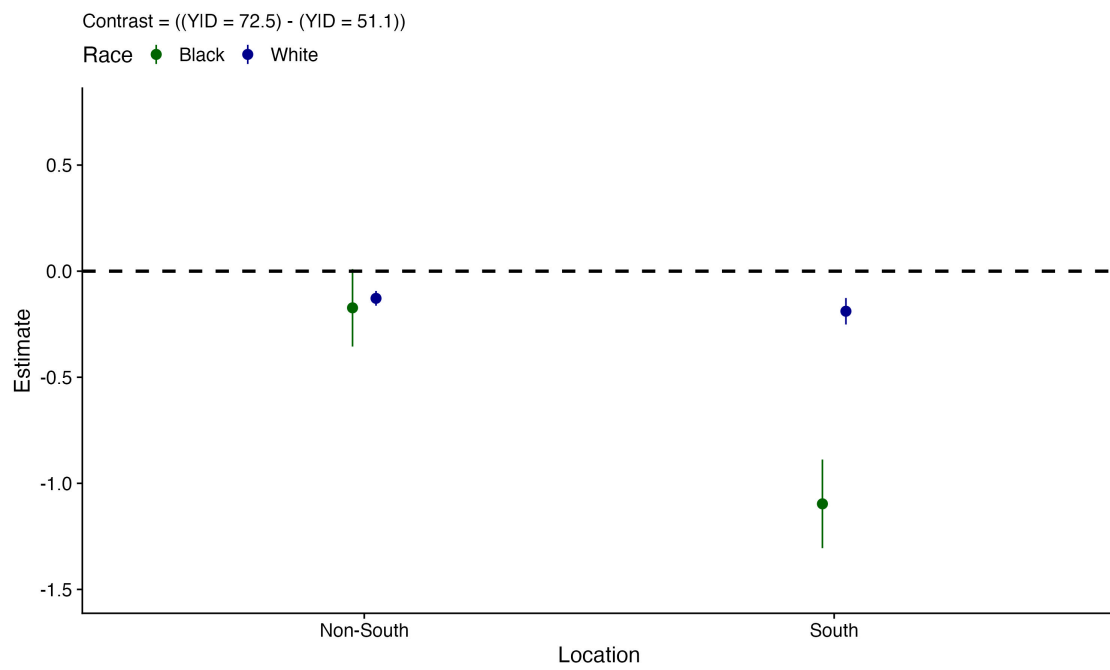


Figure 4:

Table 1: First Stage Regression of D on Instruments

	(1)
Ln(Number Governments)	7.45*** (0.29)
County Share Revenue from Federal Gov.	25.60*** (6.06)
Num.Obs.	4855
F	198.15

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001
First stage relationship includes death decade fixed effects and heteroskedasticity SEs.

Table 2: Estimates of the Association Between Segregation and Longevity

	Black	Black + C	Black + C + Period	White	White + C	White + C + Period
D	-0.082*** (0.001)	-0.084*** (0.001)	-0.004*** (0.001)	-0.067*** (0.000)	-0.067*** (0.000)	-0.001*** (0.000)
Male	-1.055*** (0.029)	-1.008*** (0.030)	-0.209*** (0.021)	-0.935*** (0.007)	-0.944*** (0.007)	-0.213*** (0.005)
Education		0.048*** (0.004)	0.007** (0.002)		0.062*** (0.001)	0.013*** (0.001)
Migrated		-0.455*** (0.037)	-0.068*** (0.018)		0.016* (0.008)	-0.092*** (0.004)
Married in 1940		0.088** (0.031)	0.073*** (0.016)		-0.059*** (0.007)	0.022*** (0.004)
Num.Obs.	108681	108681	108681	2048843	2048843	2048843
R2	0.387	0.389	0.858	0.350	0.352	0.849

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001
Heteroskedasticity Robust Standard Errors in parentheses.

Table 3: Estimates of the Effect of Segregation on Longevity

	Black	Black + C	White	White + C
D	-0.029*** (0.003)	-0.033*** (0.004)	-0.015*** (0.001)	-0.009*** (0.001)
Male		-1.009*** (0.045)		-0.955*** (0.010)
Education		0.037*** (0.005)		0.061*** (0.001)
Migrated		-0.256*** (0.039)		-0.209*** (0.008)
Married in 1940		0.127*** (0.033)		-0.017* (0.008)
Died in South		0.062 (0.055)		0.124*** (0.010)
Num.Obs.	108681	108681	2048843	2048843
R2	0.367	0.379	0.327	0.332
Birth Year	Yes	Yes	Yes	Yes
Birth State	Yes	Yes	Yes	Yes
Urban Rural Code	Yes	Yes	Yes	Yes
OCC	No	Yes	No	Yes

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$
Heteroskedasticity Robust Standard Errors in parentheses.

Table 4: Estimates of the Effect of Segregation on Longevity (By Gender)

	Black (Men)	Black (Women)	White (Men)	White (Women)
D	-0.033*** (0.006)	-0.032*** (0.006)	-0.011*** (0.001)	-0.008*** (0.001)
Education	0.056*** (0.008)	0.024*** (0.006)	0.083*** (0.002)	0.044*** (0.002)
Migrated	-0.251*** (0.061)	-0.261*** (0.050)	-0.163*** (0.013)	-0.246*** (0.011)
Married in 1940	0.121* (0.053)	0.134** (0.044)	-0.078*** (0.012)	0.061*** (0.012)
Died in South	0.004 (0.081)	0.125+ (0.074)	0.129*** (0.015)	0.118*** (0.013)
Num.Obs.	46166	62515	893437	1155406
R2	0.336	0.386	0.256	0.350
Birth Year	Yes	Yes	Yes	Yes
Birth State	Yes	Yes	Yes	Yes
Urban Rural Code	Yes	Yes	Yes	Yes
OCC	Yes	Yes	Yes	Yes

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Heteroskedasticity Robust Standard Errors in parentheses.

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Table 5: Estimates of the Effect of Segregation on Longevity (Weights)

	Black	Black + C	White	White + C
D	-0.032*** (0.004)	-0.035*** (0.005)	-0.016*** (0.001)	-0.012*** (0.001)
Male		-1.144*** (0.051)		-1.094*** (0.012)
Education		0.036*** (0.006)		0.063*** (0.002)
Migrated		-0.238*** (0.045)		-0.217*** (0.010)
Married in 1940		0.109** (0.037)		0.025* (0.010)
Died in South		0.086 (0.062)		0.108*** (0.012)
Num.Obs.	108681	108681	2048843	2048843
R2	0.383	0.396	0.366	0.373
Birth Year	Yes	Yes	Yes	Yes
Birth State	Yes	Yes	Yes	Yes
Urban Rural Code	Yes	Yes	Yes	Yes
OCC	No	Yes	No	Yes

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$
Heteroskedasticity Robust Standard Errors in parentheses.

Table 6: Estimates of the Effect of Segregation
on Longevity (Stayers)

	Black	White
D	-0.027*** (0.004)	-0.010*** (0.001)
Male	-1.008*** (0.051)	-0.951*** (0.011)
Education	0.040*** (0.006)	0.065*** (0.001)
Married in 1940	0.075* (0.038)	-0.049*** (0.010)
Died in South	-0.002 (0.049)	-0.023** (0.009)
Num.Obs.	82083	1444492
R2	0.374	0.337
Birth Year	Yes	Yes
Birth State	Yes	Yes
Urban Rural Code	Yes	Yes
OCC	Yes	Yes

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Heteroskedasticity Robust Standard Errors in parentheses. Estimates reflect the subsample of individuals who were born and died in the same county.

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